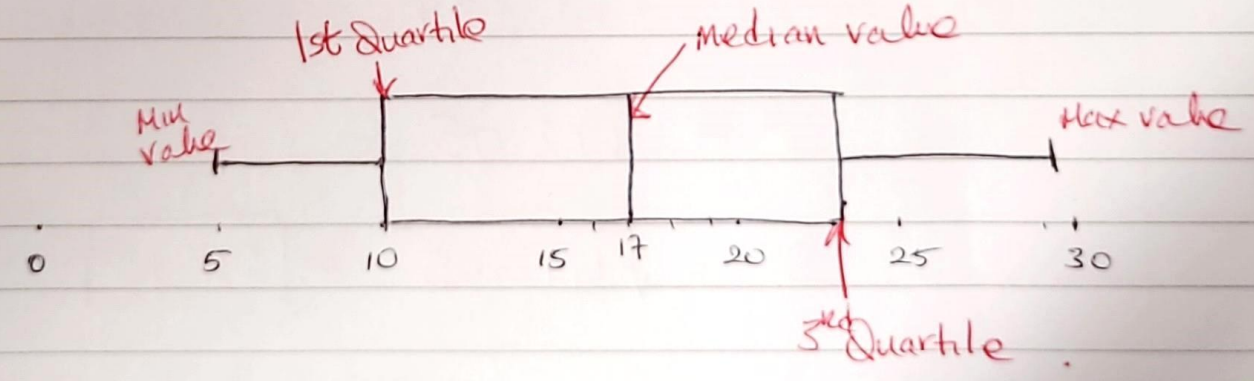
Q3 statistics & Probability

a) mean, mode, and median - give a definition of each.

All ways to measure the central tendency of a dataset.
Mode: most frequent item in a dataset
Mean: arithmetic mean / average — add up all the elements in the list and divide the answer by the number of elements in the list.
Median: sort the elements in ascending order and take the middle value

Mode generally used for categorical data (or numerical data representing categorical values). Mean is a good measure of central tendency when there are no outliers in the data. Median is a more appropriate measure to use when there are outliers/extreme values in the data.

b)



i) Median = 17

ii) Q1 = 10 Q3 = 23 interquartile range = Q3 - Q1

∴ 23 - 10 = 13

c) Grouped Data

time taken	f_i no. of students	M_i (midpoint of class interval)	f_iM_i	M_i^2	$f_iM_i^2$
0 - 20	7	10	$7\times10 = 70$	100	$7\times100 = 700$
20 - 40	16	30	$16\times30 = 480$	900	$16\times900 = 14,400$
40 - 60	11	50	$11\times50 = 550$	2500	$11\times2500 = 27,500$
60 - 80	3	70	$3\times70 = 210$	4900	$3\times4900 = 14,700$
80 - 100	3	90	$3\times90 = 270$	8100	$3\times8100 = 24,300$
	40		1580		81,600

i) Mean $\Rightarrow \bar{x} = \dfrac{\Sigma f_i M_i}{\Sigma f_i} = \dfrac{1580}{40} = \boxed{39.5}$

ii) Standard deviation: $s^2 = \dfrac{\Sigma f_i M_i^2 - M(\bar{x})^2}{M-1}$ where $M = \Sigma f_i$

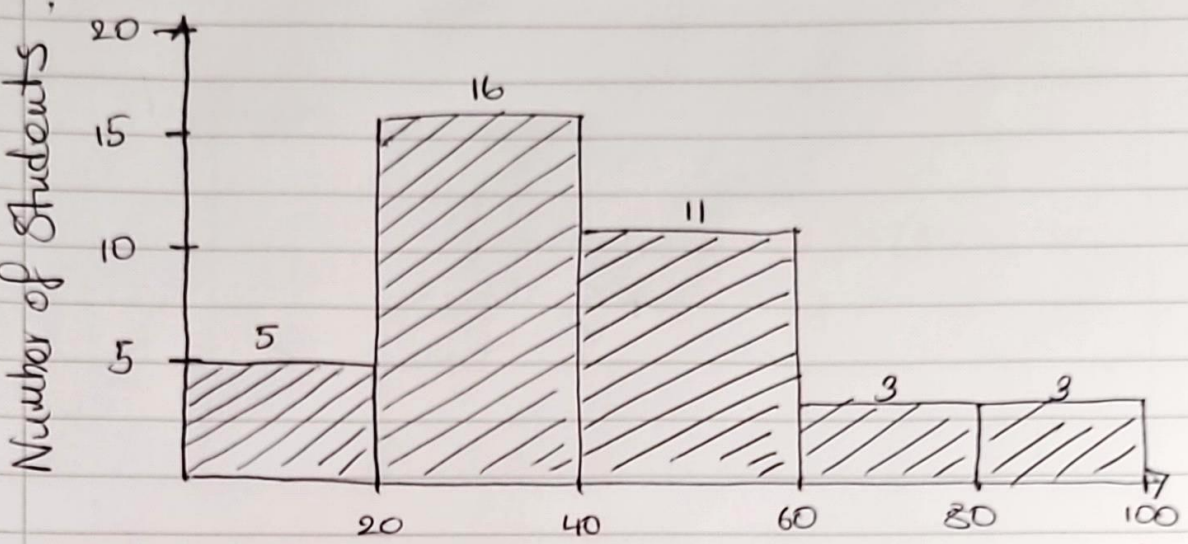
$$= \frac{81600 - 40(39.5)^2}{40-1}$$

$$= \frac{81600 - 62410}{39}$$

$$s^2 = 492.05$$

$$s = \sqrt{492.05} = \boxed{22.18}$$

iii) Histogram.



(include labels!)

iv) Comment on the symmetry or skewness + explanation.

d) Probabilities sum up to 1.

$0.02 + 0.21 + 0.17 + 0.46 + 0.04 = 0.09$

$? = 1 - 0.09$

Missing value. $= \boxed{0.1}$

e) Normal distribution $\quad a = \frac{X - \mu}{\sigma}$

$\mu = 65$

$\sigma = 9$

$X = 40$

$= \frac{40 - 65}{9} = -2.78$ (negative)

look up z table

$P[X > 40] = P[Z > -2.78] = P[Z < 2.78] = 1 - P[Z > 2.78]$

$= 1 - 0.0027 = \boxed{0.9973}$